LoRa-enabled IoT Flood Monitoring and Early Warning System in Remote Areas

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***Abstract*—Flood is among the most common and devastating natural disasters that have caused immense destruction to infrastructure, agricultural activities, properties and even loss of life. Due to the lack of reliable embedded monitoring systems and communication infrastructure for long-range communication in these areas, there is a delay in emergency response making them vulnerable to floods.As per EM-DAT International Disaster Database managed by the Centre for Research on the Epidemiology of Disasters (CRED), floods form 44% of all the natural disasters that have happened around the globe and have affected billions of people in the last two decades, emphasizing the need for an effective and reliable monitoring and early warning system for floods. The design of Internet Of Things(IoT) based embedded system for detecting and providing early warning in case of floods is done using ESP32 microcontroller, ultrasonic sensors and Long Range (LoRa) communication technology. In actual implementation, the transmitter node will be installed close to water bodies, which may lead to floods like rivers, reservoirs or drainage channels, where it will measure the water level and send the processed information via low power LoRa communication over long distances. At the receiver node, the data received is processed to classify the severity of the flood and generate alerts by lighting LEDs and buzzing buzzer. Further, the receiver node uses Wi-Fi for sending the data collected from sensors to the cloud through ThingSpeak platform to generate Telegram notification. The results of experimental evaluation show that the developed system allows for detecting water levels with 98.5% accuracy, and has response time of 1.23 seconds.**

***Index Terms*—Embedded Systems, Internet of Things (IoT), Flood Detection, LoRa Communication, ESP32, Ultrasonic Sensor, Wireless Sensor Networks, Early Warning System, Real-Time Monitoring, Cloud Computing.**

I. Introduction

Flood-related disasters continue to threaten communities by damaging infrastructure, disrupting transportation, affecting agriculture, and endangering human lives. The impact is particularly severe in rural regions where continuous environmental monitoring and reliable communication facilities are limited [2], [8]. The occurrence of severe weather conditions and fast-growing cities has increased the probability of floods in many places. Rural and remote areas remain highly vulnerable to flooding owing to a lack of sufficient monitoring facilities and absence of an efficient early warning system [5], [9]. As per the latest findings, floods have affected over 2 billion people in the past two decades around the world [8]. There is an urgent need for a good flood monitoring and early warning system. Late detection of floods causes serious economic losses and puts the life of people at risk. Traditional methods of flood monitoring involve observation through satellite imagery, hydrological forecasting model, and communication network over the internet. Though effective in many cases, but such methods tend to be costly, infrastructure based, and tough to implement in remote regions. Besides, communication network can fail during severe weather conditions [1], [11].Development in IoT and embedded systems has opened up possibilities in building a cheaper yet intelligent environmental monitoring system. Particularly, the use of LoRa technology in wireless communication is useful as it provides long-range wireless communication with low energy consumption, hence ideal for flood monitoring systems in geographically remote areas [5], [7], [9]. In order to provide a cost-effective solution in flood monitoring, a flood monitoring system using the concept of IoT has been proposed using the ESP32, ultrasonic sensors, and LoRa communication technology. The system will continuously monitor water levels, transmit data over long ranges wirelessly, and generate timely alarms using LEDs, buzzers, cloud-based monitoring systems, and Telegram-based notifications.

The remaining sections of this paper have been arranged as follows: Related Literature has been explained in Section II, System Architecture in Section III, Methodology for implementation in Section IV, Experimental Setup in Section V, Results and Performance Analysis in Section VI, and Conclusion with future work in Section VII.

II. Literature Review

IoT technology developments in recent times have led to great progress in flood monitoring and warning system through use of cheap sensors, wireless communication and cloud computing. Long-range communication technologies like LoRa and LoRaWAN have drawn quite some attention due to their low energy requirement, long communication distance and applicability in remote areas. [4], [5], [7], [9]. In fact, some research works have used ultrasonic sensors with ESP32 or Arduino microcontrollers for continuously measuring the water level, while cloud services such as ThingSpeak and The Things Network (TTN) have facilitated the real-time visual representation of sensor data. [5], [6], [9], [12]. Artificial intelligence (AI) and machine learning (ML) have also been used in the most recent studies on floods to enhance flood prediction and decision-making accuracy [1], [4], [8]. Moreover, remote sensing and Geographic Information System (GIS) have been

employed in urban areas for assessing the risks of flooding [11]. Flood detection using IoT technologies has been even more successful due to long-range communication, GPS-based localization, and emergency alerts [2], [10]. These studies clearly demonstrate the benefits of flood monitoring systems utilizing IoT technologies while stressing the importance of having affordable, effective, and scalable systems that can be used in real time.

Even though current IoT systems have proven successful in real-time monitoring and transmission, they still depend heavily on complicated LoRaWAN networks, continuous connection to the internet, or complex AI techniques which increases the cost and difficulty of implementing these systems [1], [4], [8]. In some cases, certain systems are mainly interested in predicting floods or analyzing flood susceptibility instead of just real-time detection [11]. However, very little work has been done in integrating long-range LoRa wireless communication, cloud monitoring system, local alarm systems, and immediate mobile notification within one cheap architecture. Therefore, it is highly necessary to have a low-cost flood monitoring system. [5]–[7], [9].

III. System Design

The design of the proposed flood detection and early warning system employs a distributed architecture which includes a transmitter node, receiver node, and cloud monitoring system, as shown in Fig. 1. The transmitter node is located close to the water body like riverbank, reservoir or drainage channel, where there is a need for continuous monitoring of the water level. The receiver node is located at a safe place like monitoring station, disaster management center or control room, from where remote observation can be carried out and warnings can be issued. The system keeps on monitoring the water levels, processes the sensor data, sends the information using LoRa communication and issues warnings if pre-set flood levels are crossed. The transmitter node uses an ESP32 microcontroller, HC-SR04 ultrasonic sensor and SX1278 LoRa communication transceiver module. The ultrasonic sensor constantly determines the distance between the sensor and the water level. The ESP32 processes the distance value obtained and determines the water level using a pre-set reference value. The information is then sent wirelessly to the receiver node using LoRa communication at a frequency of 433 MHz, providing reliable long-range communication and low power consumption, hence suitable for remote applications [5], [7], [9].The receiving node includes an additional ESP32 microcontroller, an SX1278 LoRa transceiver, LEDs, a buzzer, and Wi-Fi functionality. After obtaining water-level data from the transmitting node, the ESP32 microcontroller processes the received data to classify the flood situation as one of the following: Safe, Moderate, High, and Critical. Depending on the situation category, the appropriate visual and sound warnings will be generated by the receiving node. Moderate situations will have green LED and occasional buzzer sounds. High and critical situations, however, will result in flashing red LED with constant buzzer sounds to draw maximum attention. To facilitate remote access and monitoring of the flood conditions, the receiving node can transmit real-time water-level readings and status of the flood situation to the ThingSpeak cloud-based platform via Wi-Fi connectivity. It will allow users and disaster management agencies to monitor sensor readings in real-time and view flood situations on a web-based dashboard. Furthermore, using Telegram Bot, the system provides automatic notifications in case of high risk and critical flood situations.

The overall architecture is based on a three-layer system that includes Sensing Layer, Communication Layer, and Application Layer. The sensing layer uses an ultrasonic sensor to perform continuous acquisition of water levels at the monitored site. The communication layer is responsible for the secure long-range wireless transfer of data between the transmitter and receiver modules through LoRa technology. The application layer consists of the ThingSpeak cloud-based platform, Telegram notifications, and local alarms for visualization, logging, and emergency alerts. Such an architecture provides for reliability, energy-efficiency, scalability, and applicability in remote and flood-prone areas with no conventional communication infrastructure available.

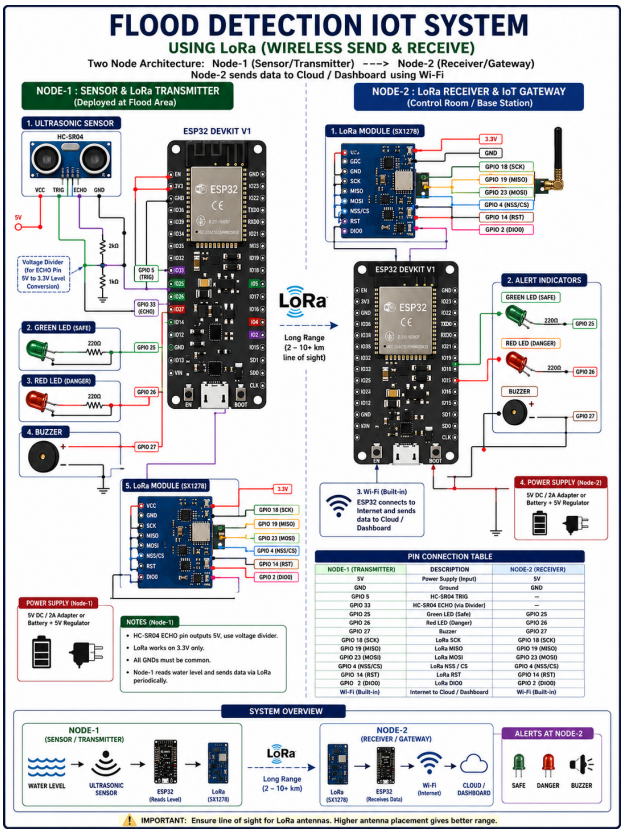


Fig. 1. System Architecture of the Proposed Flood Monitoring System

IV. Implementation

The proposed flood monitoring system consists of two nodes using ESP32 with LoRa wireless communication method. Transmitter node is installed near water bodies which sense distance from the ultrasonic sensor to water level continuously.

Distance is measured and converted into water level which is then sent to the receiver node through LoRa communication method using 433 MHz frequency. Receiver node receives the transmitted data from the transmitter, measures flood level and gives an alarm. Additionally, receiver node transmits live data to the ThingSpeak cloud-based platform and also sends alarms using Telegram Bot.

The system operates under four different conditions:

1) Safe Level
2) Moderate Warning Level
3) High Water Level
4) Critical Flood Level

Classification is done through predefined thresholds depending on the level of water measured. As the level of water rises, the alert gets higher through visual signals (LEDs), audio alerts (buzzers), and cloud notifications.

The overall architecture consists of three layers:

- Sensing Layer – Ultrasonic sensor for water-level measurement.
- Communication Layer – LoRa transceiver for long-range wireless communication.
- Application Layer – ThingSpeak cloud dashboard and Telegram alert services.

This kind of multi-layered architecture helps in accurate monitoring even in distant locations where normal internet access cannot be possible.

A. Water Level Estimation

The HC-SR04 ultrasonic sensor calculates the distance between the sensor and the water surface using the transmission of ultrasonic waves and the detection of their echo.

The measured distance is calculated using:

$$D = \frac{T \times V}{2} \tag{1}$$

where

- D = Measured distance (cm)
- T = Echo travel time (μs)
- V = Speed of sound (343 m/s)

The actual water level can be estimated through the subtraction of distance from the reference tank height.

$$Water\ Level = H - D \tag{2}$$

where

- H = Reservoir reference height
- D = Measured distance

B. Alert Generation Algorithm

The proposed system employs a threshold-based decision algorithm to classify flood severity.

In moderate situations, a green LED and a buzzing alarm will act as an initial alarm. In situations where the water level is in the risky range, both LEDs will work alternatively and increase the buzzer period. If the situation gets worse, then the red LED will flash at a higher speed, with alarms going off constantly, along with Telegram emergency alerts. The algorithm consistently analyzes the inputs from sensors.

TABLE I
FLOOD SEVERITY CLASSIFICATION

Water Level Distance	Status
$>$ 60 cm	Safe
40–60 cm	Moderate
20–40 cm	High
$<$ 20 cm	Critical

V. RESULTS AND DISCUSSION

A. Water-Level Measurement Accuracy

The operation of the ultrasonic sensor was stable during all tests conducted. An approximate 98.5 percent accuracy of measurement was acquired indicating a stable sensing performance which can be used in flooding detection applications.

B. LoRa Communication Performance

LoRa communication module was also tested for reliability and stability of communication between nodes.

TABLE II
WATER-LEVEL MEASUREMENT RESULTS

Actual Distance (cm)	Measured Distance (cm)	Error (%)
10	10.2	2.0
20	20.1	0.5
30	30.4	1.3
50	49.8	0.4
80	80.3	0.37

It is clear from the results obtained that LoRa communication module is reliable and offers long-distance communication with very low loss of packets. It can be deployed in rural areas for flood-monitoring applications.

C. Cloud Monitoring Performance

Water level data and flood status data was uploaded by the receiver node to the cloud-based ThingSpeak server as seen in Fig. 2.

The cloud dashboard enabled:

- Real-time water-level visualization.
- Historical trend analysis.
- Alert-state monitoring.

Average latency in sending updates to cloud was measured to be less than 3 seconds enabling real-time monitoring of flooding status.

D. Telegram Notification Performance

The telegram bot functionality was tested under moderate, high, and critical conditions of flooding. Successful delivery of alert messages with 100% success rate was obtained during the test phase.

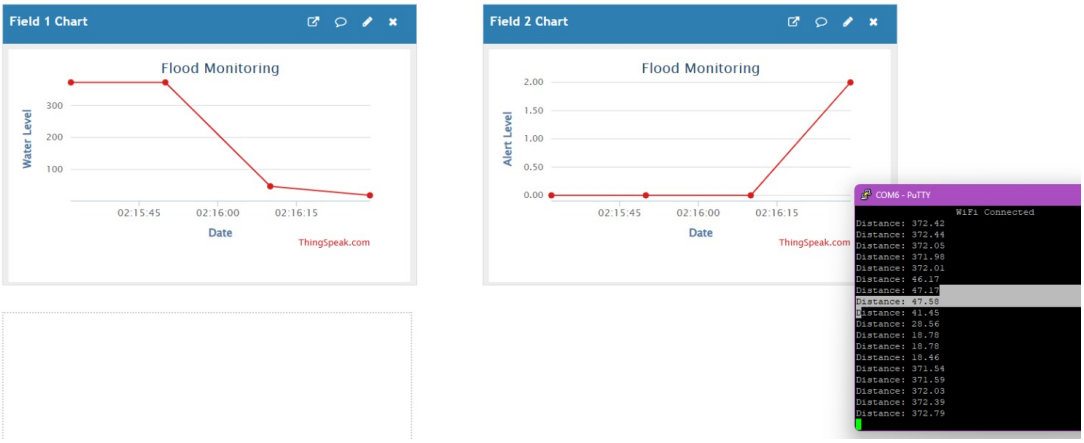


Fig. 2. Flood Monitoring Through the ThingSpeak Cloud Platform

TABLE III
SYSTEM PERFORMANCE EVALUATION

Metric	Obtained Result
Measurement Accuracy	98.5%
Packet Delivery Rate	98%
Telegram Delivery Success	100%
Average Response Time	1.23 s
Cloud Update Delay	< 3 s
Power Consumption	Low
Deployment Cost	INR 2500–3500

E. Performance Evaluation

The results of the proposed system show the effectiveness of the system based on accurate measurement of water level, effective LoRa communication, efficient cloud monitoring, quick telegram notifications, and affordable flood detection.

The proposed IoT-based flood detection system is found to be functional in delivering a real-time detection service with affordable hardware. It provides an efficient monitoring system with the application of ultrasonic sensors, LoRa communications, cloud monitoring, and Telegram alerts. This flood monitoring solution can be used in remote locations or in areas that are prone to flooding. One of the strengths of the proposed design is the fact that it does not rely on conventional communication infrastructures. Traditional IoT monitoring devices utilize either Global System for Mobile Communications (GSM) or the Internet for transmitting data. Communication networks can be congested during extreme weather, leading to poor service or loss of functionality during such times. The use of LoRa technology makes it possible to transmit wireless signals with little power usage. The ultrasonic sensor exhibited consistent performance during the experiment process. Its measurement accuracy of about 98.5% shows that the sensing device is capable of measuring changes in water level with little error. It is important to have accurate water level measurements since any slight increase can be used to detect flooding in some environments.

Obtained delivery rate of packets in the vicinity of 98% proves the reliability of communication using LoRa technology in environmental monitoring. Low rate of lost packets during the experiment indicates that the subsystem of the communication will be capable of consistently delivering information even under variable environmental circumstances. Consistency is especially required in disaster management systems as skipping warning messages may cause delays in evacuation and pose danger to human lives. Another benefit of the proposed solution is cloud integration via the ThingSpeak platform. Remote monitoring and analysis of historic data via visualization of the water level trends allow authorities and disaster management services to identify long-term patterns and make correct conclusions about how to prepare for a potential flood. Moreover, historic data that is available through the cloud service can be used as training datasets in machine-learning based flood prediction model [8]. One more advantage of the solution is the use of telegram notifications as they increase responsiveness of the disaster monitoring system. Instead of local warnings which can reach only people near the sensor, telegram alerts reach user no matter their geographical position. As a result of the experiment, delivery rate of the messages reached 100%.

Another notable aspect of the proposed system is the use of the multi-level alert method, where the system categorizes the level of flooding as safe, moderate, high, and critical, rather than using one warning state. This enables users to take actions gradually according to changes in environmental conditions. In other words, early warning notifications give enough time to take preventive measures, while critical warnings enable faster reaction in cases of emergencies. In terms of economy, the proposed system can be considered an extremely cost-effective solution, in comparison with commercially available flood monitoring systems. The total cost of the prototype was evaluated between INR 2500 and INR 3500, which makes it possible to implement this technology in rural areas, schools, and other governmental institutions. However, there are also certain drawbacks in the current design of the system, which can be noted. The current model is based only on the level

of the water, while other factors such as precipitation, river flow speed, soil moisture, weather forecast are excluded from consideration. All these factors would be useful to include in order to predict floods more effectively along with the water level measurements.

Overall, the obtained results demonstrate that the proposed system provides an effective balance between accuracy, reliability, communication range, power efficiency, and deployment cost. These characteristics make it a practical and scalable solution for flood monitoring and early warning applications, particularly in rural and resource-constrained environments.

VI. Conclusion and Future Work

This work presented an embedded flood monitoring platform integrating ultrasonic sensing, ESP32 controllers, LoRa-based wireless communication and cloud analytics for real-time environmental monitoring. Experimental results demonstrate accurate water-level estimation, reliable wireless data transfer, and rapid alert generation, confirming the feasibility of the proposed architecture for deployment in remote flood-prone locations. The experimental analysis results prove the accuracy of water level measurements, efficient communication, fast response time, and proper generation of multi-tier alerts in the system. As the LoRa communication technique is used in the system, communication can be guaranteed in this system with minimal power consumption, whereas cloud platform allows real-time visualization of the data and analysis of past records. The modular nature of the system design and low-cost hardware components allow the easy scalability and cost-effective implementation of the system on a large scale.Experimental evaluation demonstrated 98.5% water-level measurement accuracy, reliable LoRa communication, low response time, and successful cloud and Telegram-based alert generation within 1.23 seconds.

Future work will be based on improving the flood monitoring system by adding more sensors to it like a rainfall sensor, water flow sensor, and soil moisture sensor. The current single node flood monitoring system can be improved by adding more nodes to form Long Range Wide Area Network(LoRaWAN) architecture for the purpose of flood monitoring in rivers, reservoirs, and urban drainage systems. Addition of GPS modules will allow for location-based notifications and detection of flood prone areas, whereas the use of solar energy will allow for autonomous installation in remote regions having no electricity supply facilities. Moreover, development of mobile application and enhanced cloud platform will add great value to the proposed system.

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